

CS3 - Case Study Rubric

Due: TBD

Submission format:

- Upload link to GitHub repository to Canvas

Individual Assignment

General Description: Submit to Canvas a link to your case study repository, which should include your code, appropriate visualizations, and a brief summary of your results.

Why am I doing this? With your background knowledge in time series analysis, you are now ready to apply those skills to more advanced time series techniques, as you encounter a real-world scenario with real data and stakeholders. As you work through this case study, you will gain experience in time series modeling, visualizing seasonality, ACF/PACF plots, and forecasting. This will demonstrate how time series analysis can be used to forecast future data accurately and comprehensively, which will prepare you to apply these concepts to future projects.

What am I going to do? You can explore the project repository to find articles and documentation to help you get started, as well as the raw data file you will be using. You will use this to build a SARIMAX model, if appropriate, to forecast the 2023 temperature measurements one year in advance and provide a visualized confidence interval of said forecast. Your deliverable will include:

- GitHub repository containing all materials, code, and conclusions for this case study
 - Written PDF briefly summarizing your conclusions about precipitation's effect on temperature and your final model and final forecast
 - Scripts with your EDA, ACF/PACF plots, model diagnostics, and final model forecast
 - A JPG image of your final visualized forecast and confidence interval
 - An updated README file containing information about your data and the contents of your repository

How will I know I have succeeded? You will meet expectations when you follow the criteria in the rubric below.

Formatting	• GitHub repository containing all materials ◦ Submit a link to the repo ◦ Contents: ■ Data and code used for all portions of the analysis • Scripts: EDA, Analysis ■ Written summary of results - PDF • Including References - IEEE citation style ■ Visualization of forecast - JPG

README.md	● <u>Goal:</u> Brief summary of what the project is – does not need to be detailed but should provide enough information to orient people to the subject matter and contents of the repository. ○ Document file mapping so a viewer can easily navigate ○ Provide instructions for reproducibility
Scripts	● <u>Goal:</u> Annotated scripts containing all of the code used to create and execute your time series model, forecast, and modifications to the model. ○ Use the dataset provided to you in the GitHub for this assignment called NYC_temperature.csv ○ Include comments so a viewer can easily comprehend and reproduce your process ○ You should have two scripts: EDA and Analysis ■ EDA: preprocess the data to aggregate monthly averages and limit data to 1970-2023, conduct EDA to explore any possible seasonality component in the temperature data and decide whether or not differencing is needed ■ Analysis: create ACF and PACF plots to determine candidate SARIMAX models (or ARIMAX if no seasonality component was found) with precipitation as the exogenous variable, fit SARIMAX models and determine the best final model, determine significance of precipitation in final model and remove to fit SARIMA/ARIMA model if applicable
Written summary	● <u>Goal:</u> Provide brief conclusions and summarize your methods and results. ○ Include final model ○ Include p-value and accompanying conclusions about the significance of precipitation in the model ○ Include a brief reflection on the impacts of your discoveries
Visualization	● <u>Goal:</u> Provide a clear forecast for NYC 2023 temperature measures. ○ Can be output from your Analysis script, but should be uploaded separately as a JPG ○ Should appear clear and comprehensible to non-technical viewers ○ Should graph the 2023 forecast as well as shading to represent a 95% confidence interval
References	● <u>Goal:</u> Cite any resources referenced in your project. ○ Use IEEE citation style